SAEFRig: An Automatic Rigging Model Based on Structure-Aware Encoding and Fusion

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***Abstract*—Automatic skeletal rigging is a crucial task in 3D animation, aiming to generate skeletal structures and skinning weights for three-dimensional character models to enable realistic motion control. However, existing methods typically rely on single feed-forward predictions, making it difficult to establish robust shape-skeleton correspondences under diverse shapes and non-standard poses, leading to unstable and inconsistent results. To address this, this paper proposes SAEFRig, a three-stage automatic rigging framework. The method transforms unordered point clouds into structurally ordered representations via Structure-Aware Encoding (SAE), and employs a Multi-Skeleton Fusion (MSF) module to integrate complementary information from skeleton estimation and skinning weight prediction, progressively refining the rigging outcome. Experiments conducted on datasets such as Mixamo and VRoid (including their extended versions Mixamo+ and VRoid+) demonstrate that SAEFRig outperforms existing methods in both skeleton generation and skinning weight prediction, exhibiting stronger robustness and accuracy especially under non-standard poses.**

***Index Terms*—Automatic Rigging, Skeleton Estimation, Skeleton-Aware Positional Encoding**

I. Introduction

Automatic rigging [1], [2] is a pivotal task in 3D computer animation, which aims to construct skeletal structures and define skinning weights for 3D character models, thereby enabling realistic and expressive kinematic motion. The core challenge of this process lies in accurately establishing a robust structural-semantic correspondence between the model's surface geometry and its underlying kinematic skeleton. This challenge is exacerbated when handling characters with diverse shapes and non-standard poses, resulting in unstable and implausible rigging outcomes.

In recent years, deep learning has become the dominant paradigm for automatic rigging [3]. Current methods generally follow one of two strategies: template-based approaches [3], [4], which rely on predefined skeleton templates and predict joint positions and skinning weights within a fixed topology, and template-free methods [1], which directly infer joint locations, connectivity, and skinning weights to form a skeleton from scratch. Despite their advances, a key limitation persists across both paradigms: most methods operate in a single feed-forward pass, taking a raw point cloud as input and predicting rigging parameters directly. This single-pass design, however, introduces two inherent limitations. First, the predictions of point-cloud networks are highly sensitive to the spatial distribution of the input points, often causing inconsistent rigging results for the same character under different poses. Second, the scheme fails to explicitly model the intrinsic, complementary relationship between skeletal joints and skinning weights, thereby preventing an iterative refinement that could improve the coherence and accuracy of the final rigging. Consequently, the rigging quality is highly dependent on the initial pose and local geometry, often leading to artifacts when the input deviates from the training distribution. Thus, establishing robust, pose-invariant shape–skeleton correspondences remains a critical challenge, especially when processing the diverse outputs of AIGC pipelines [5].

To address these limitations, we propose SAEFRig, a novel automatic rigging framework designed to progressively refine the rigging results through Structure-Aware Encoding (SAE) and Multi-Skeleton Fusion (MSF). The novelty of our approach lies in two key aspects. First, we introduce structure-aware encoding that leverages an initially estimated skeletal hierarchy and skinning weights to embed each point within an ordered structural context. This transformation enables the network to more accurately model the correspondence between the surface geometry and the skeleton. Second, the multi-skeleton fusion harnesses the complementary information between joint localization and skinning semantics to generate multiple estimations of the skeleton, which are then integrated by a fusion network to boost the overall accuracy and robustness of the final output. This progressive refinement strategy is fundamentally different from the single-pass paradigm, as it explicitly allows the network to correct initial estimations and reinforce consistency, effectively addressing the limitations of joint offset and poor pose generalization in current approaches.

Extensive experiments on the standard benchmarks, including Mixamo [6] and VRoid [7], demonstrate that SAEFRig achieves state-of-the-art performance in both skeleton generation and skinning weight prediction. The results significantly outperform existing methods, particularly on models with complex topologies and diverse poses. Our work demonstrates the potential of incorporating structural reasoning and iterative refinement into the automatic rigging pipeline, offering a more robust solution for AIGC-generated content, and is thoroughly validated across various scenarios.

II. Related Works

A. Geometric Skeleton Extraction

Early skeleton extraction methods primarily relied on geometric analysis [8], [9]. While these geometry-driven methods are applicable to a variety of character shapes, they often produce skeletons that are either overly complex or too simplified, lacking the structural integrity required for expressive character animation. To generate animation-ready skeletons,

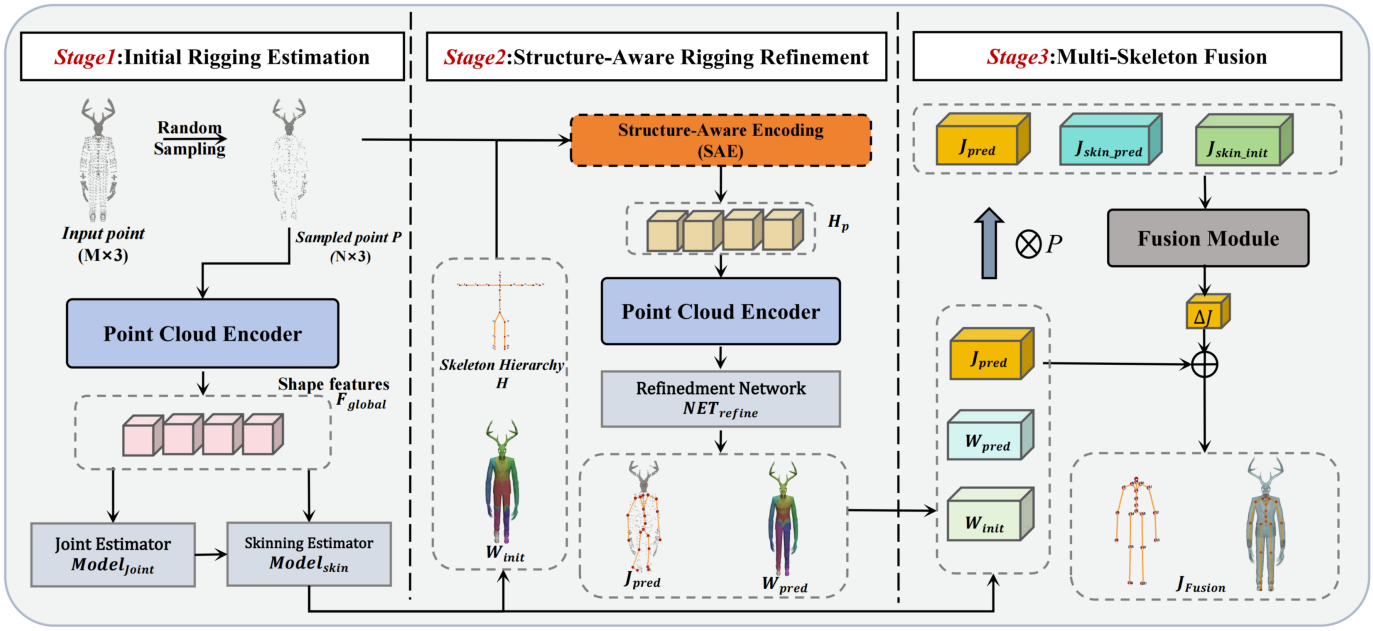

Fig. 1: **Overview of the proposed framework**. Our three-stage pipeline estimates rigging from a 3D character. Starting from a point cloud, Stage1 predicts initial skinning weights. Stage2 refines joints and weights via structure-aware encoding to enhance pose robustness. Finally, Stage3 fuses multiple skeleton estimations to output the final, high-consistency rigging result.

later works [10] incorporated shape priors or template information. For instance, Pinocchio [11] employed fixed template, optimizing vertex positions to fit the target model. Motion-guided methods [12] also emerged, using motion data to segment shapes, then estimate kinematic skeletons. Although these methods greatly improved usability, they still struggled with models exhibiting significant topological variations or diverse poses.

B. Skinning Estimation

Skinning estimation [13], [14] assigns weights to each vertex of a 3D model, defining its deformation in response to skeletal movement. Early geometric methods [15], [16] calculated these weights based on distances between joints and vertices, which often failed to capture semantic deformation on complex models. Recently, learning-based methods, particularly those using Graph Neural Networks (GNNs), have become dominant. These approaches [17]–[19] model the spatial relationships between the mesh and the skeleton to predict skinning weights. However, their performance heavily depends on the consistency of character pose and skeleton topology with the training distribution, limiting their robustness in practice.

C. Learning-based Rigging

Learning-based rigging methods can be categorized by their use of skeletal topology. Template-free methods predict joint distributions and construct topology adaptively [1], [20], offering flexibility but complicating motion retargeting. Template-based methods predict parameters of a fixed skeleton [3], [21], ensuring stability but limiting shape generality.

To animate the growing variety of AI-generated characters, recent works aim to unify rigging for arbitrary shapes and poses via large-scale data and powerful generative models. Common strategies include constructing massive datasets [2], [22] and employing autoregressive or other generative architectures to predict both skeleton and skinning [4], [23]. Nevertheless, a key remaining challenge for these data-driven methods is their sensitivity to variations in global pose and body proportions, which hampers reliable generalization. Still, these data-driven methods tend to be sensitive to changes in global pose and body shape, leading to performance drops when generalizing across diverse characters. In contrast, our method explicitly addresses this issue by enriching kinematic features with semantic perceptual cues, which helps achieve more stable predictions even under complex poses.

III. Method

A. Overview

Given a 3D character model, automatic rigging aims to produce the skeleton and associated skinning weights to derive its motion animations. In this work, we deal with humanoid characters represented as point clouds. This approach assumes a fixed hierarchy structure of the 22-joint skeleton as the template, and trains the network to predict both the 3D coordinates of the skeleton joints $\mathbf{J} \in \mathbb{R}^{K\times 3}$ and the skinning weight matrix $\mathbf{W} \in \mathbb{R}^{M\times K}$, where M and J denote the vertex number and joint number respectively.

We propose a three-stage automatic rigging pipeline, as illustrated in Fig. 1. The first stage performs an initial rigging estimation, where we sample a point cloud from the original

point clouds and feed it into the network to estimate the skeleton and skinning weights. The second stage performs structure-aware rigging refinement, which leverages the initial estimation to compute the Structure-Aware Encoding to enhance the relationship between the shape and its skeleton. It produces the refined estimation of skeleton joints and skinning weights. Finally, in the third stage, we use a Multi-Skeleton Fusion module to fuse the information from the previous stages and output the final rigging results.

B. Stage 1: Initial Rigging Estimation

The first stage inputs the sampled point cloud $\mathbf{P} \in \mathbb{R}^{N\times 3}$ to the Point Cloud Encoder to extract shape features. These shape features are then separately fed into a joint network and a skinning network to ultimately output an initial estimation of the skinning weights $\mathbf{W}_{init} \in \mathbb{R}^{N\times K}$.

Specifically, the sampled point cloud $\mathbf{P}$ is fed into the Point Cloud Encoder to extract a global shape feature F_{global}. The encoder typically consists of a stack of self-attention and MLP layers that hierarchically aggregate local and global information.

Next, we predict the skeleton joints and skinning weights with Joint Estimator and Skinning Estimator. For skeleton joints, the network takes the feature F_{global} as input and predicts the joint positions $J_{init} \in \mathbb{R}^{K\times 3}$:

$$J_{init} = \mathrm{Model}_{joint}(F_{global}). \tag{1}$$

For the skinning weights, the Skinning Estimator concatenate the point coordinates and the repeated global feature, and leverage an MLP (named MLP_p) to compute the point features F_p. Then another MLP network (named MLP_j) is employed to compute the joint features F_j based on the global feature. These two feature matrices are multiplied to estimate the skinning weight matrix W_{init}:

$$F_p = \mathrm{MLP}_p\left(\mathrm{concat}\left(P, F_{global}\right)\right) \tag{2}$$

$$F_j = \mathrm{MLP}_j\left(\mathrm{concat}\left(J_{init}, F_{global}\right)\right) \tag{3}$$

$$W_{init} = \mathrm{Model}_{skin}\left(\mathbf{P}, \mathbf{F}_{\text{global}}, \mathbf{J}_{\text{init}}\right) = \mathrm{softmax}\left(F_j F_p^T\right). \tag{4}$$

Although the first stage predicts both skeleton joints J_{init} and skinning weights W_{init}, but in our implementation, only the skinning weights from the first stage are used in the following process.

C. Stage 2: Structure-Aware Rigging Refinement

The second stage leverages the initial skinning weights as semantic structural guidance to perform structure-aware rigging refinement. The motivation is that skinning weights help align the distributions between the character's surface geometry and its underlying skeleton. Inspired by positional encoding in Transformers, we incorporate the skeleton hierarchy as a structural prior and bind this information to the point cloud using the associated skinning weights, thereby forming the Structure-Aware Encoding (SAE) module. It transforms the unordered point cloud into a sequential, structure-aware representation that explicitly captures the correspondence between points and joints, significantly enhancing the network's capacity for rigging refinement.

In our implementation, the skeleton hierarchy is serialized using a depth-first search (DFS) order and attached to the point cloud sampled from the character surface via the skinning weights:

$$H_P = [P, W_{init}H] \tag{5}$$

where $H_P \in \mathbb{R}^{N\times 4}$ denotes the sequential representation of the point cloud, $W_{init} \in \mathbb{R}^{N\times K}$ represents the skinning weight matrix, and $H \in \mathbb{R}^{K\times 1}$ stands for the DFS-ordered indices of the skeleton joints.

The point clouds, along with its Structure-Aware Encoding, is then fed into the Refinedment Network:

$$J_{pred}, W_{pred} = \mathrm{NET}_{refine}(H_P). \tag{6}$$

This network employs a multi-head attention mechanism in the encoder to extract shape features, which are progressively used to predict the refined skeleton joints J_{pred} and skinning weights W_{pred}. The inclusion of structure-aware encoding H_P enables the network to maintain robust performance for non-standard poses by establishing explicit shape-skeleton correspondences beyond mere spatial point coordinates.

D. Stage 3: Multi-Skeleton Fusion

So far we have obtained the rigging estimations from the previous two stages, i.e. the initial skinning weights W_{init}, the refined skinning weights W_{pred}, and the refined skeleton joints J_{pred}. Given the intertwined relationship between joint positions and skinning weights, this leads to their alignment by taking them as three versions of skeleton estimations. It forms the Multi-Skeleton Fusion (MSF) module as follows.

Specifically, given the skinning weights, we can regress their corresponding skeleton joint localization based on the spatial coordinates of the input point clouds:

$$J_{skin_init} = W_{init}P, \quad J_{skin_pred} = W_{pred}P \tag{7}$$

Thus, we obtain the three distinct estimations of the skeleton joint positions, i.e. J_{skin_init}, J_{skin_pred}, J_{pred}. Then we adopt the Fusion Module (implemented as a shared MLP, denoted as $\mathrm{FusionMLP}$) to fuse the multiple estimations. Specifically, we use all three estimations to predict the displacements of each skeleton joint to further refine the coherence of the skeleton and skinning weights:

$$\Delta J = \mathrm{FusionMLP}\left(\mathrm{concat}\left(J_{skin_init}, J_{skin_pred}, J_{pred}\right)\right) \tag{8}$$

$$J_{\text{fused}} = J_{pred} + \Delta J \tag{9}$$

where $\Delta J \in \mathbb{R}^{K\times 3}$ denotes the predicted joint displacement matrix, and FusionMLP denotes the dedicated MLP for the feature fusion. Finally, the predicted skinning weights W_{pred} and fused skeleton joints J_{fused} serve as the final rigging results for the target shape.

TABLE I: Comparison with RigNet and UniRig. Quantitative comparison on Mixamo, VRoid, and their pose-augmented variants (Mixamo+ and VRoid+). We report CD-J2J, CD-J2B, CD-B2B and Skin-L1 (lower is better). Our method with UniRig backbone achieves the best performance across all metrics and datasets.

	Mixamo				VRoid				Mixamo+				VRoid+			
	CD-J2J	CD-J2B	CD-B2B	Skin-L1	CD-J2J	CD-J2B	CD-B2B	Skin-L1	CD-J2J	CD-J2B	CD-B2B	Skin-L1	CD-J2J	CD-J2B	CD-B2B	Skin-L1
RigNet	0.1022	0.0886	0.0205	0.0908	0.2405	0.1007	0.0676	0.0489	0.2484	0.1093	0.0605	0.1073	0.2171	0.0934	0.0714	0.1292
UniRig	0.0116	0.0080	0.0069	0.0165	0.0044	0.0036	0.0034	**0.0047**	0.0298	0.0247	0.0221	0.0335	0.0161	0.0130	0.0115	0.0288
Ours(PCT)	0.0241	0.0178	0.0162	0.0208	0.0048	0.0030	0.0027	0.0101	0.0653	0.0539	0.0461	0.0664	0.0184	0.0144	0.0124	0.0315
Ours(UniRig)	**0.0085**	**0.0058**	**0.0049**	**0.0152**	**0.0034**	**0.0027**	**0.0021**	**0.0047**	**0.0236**	**0.0199**	**0.0171**	**0.0304**	**0.0090**	**0.0073**	**0.0067**	**0.0251**

TABLE II: Ablation study of SAE and MSF modules. We compare four configurations: baseline, w/o SAE, w/o MSF, and full model (Ours). Note that MSF does not produce new skinning weights (marked as –). Our full model achieves the best performance, demonstrating the complementary benefits of both modules.

	Mixamo				VRoid				Mixamo+				VRoid+			
	CD-J2J	CD-J2B	CD-B2B	Skin-L1	CD-J2J	CD-J2B	CD-B2B	Skin-L1	CD-J2J	CD-J2B	CD-B2B	Skin-L1	CD-J2J	CD-J2B	CD-B2B	Skin-L1
baseline	0.0116	0.0080	0.0069	0.0165	0.0044	0.0036	0.0034	**0.0047**	0.0298	0.0247	0.0221	0.0335	0.0161	0.0130	0.0115	0.0288
w/o SAE	0.0097	0.0064	0.0053	0.0160	0.0035	**0.0027**	**0.0021**	**0.0047**	0.0262	0.0222	0.0196	0.0324	0.0145	0.0120	0.0107	0.0300
w/o MSF	0.0107	0.0073	0.0064	–	0.0044	0.0036	0.0034	–	0.0253	0.0213	0.0187	–	0.0110	0.0090	0.0087	–
Ours	**0.0085**	**0.0058**	**0.0049**	**0.0152**	**0.0034**	**0.0027**	**0.0021**	**0.0047**	**0.0236**	**0.0199**	**0.0171**	**0.0304**	**0.0090**	**0.0073**	**0.0067**	**0.0251**

IV. Experiments

A. Experiment Settings

Datasets. We conduct evaluation experiments on Mixamo [6] and VRoid [7] datasets. Mixamo dataset features diverse character shapes with a limited set of poses, while VRoid dataset consists of uniform humanoid models all in a standard pose (i.e., T-pose). To validate the performance on non-standard poses, we construct extended datasets named Mixamo+ and VRoid+. That is, we extracted 5,120 frames from 10 distinct animations [6] and randomly applied the poses to the rigged characters in the original datasets.

Metrics. We evaluate both skeletal structures and skinning weights. For skeletal predictions, we use three distinct Chamfer Distance-based metrics: CD-J2J measures the accuracy of joint placement by computing the bidirectional distance between predicted and ground-truth joint positions; CD-J2B assesses how well predicted joints align with the true bone structures by calculating the distance from joints to corresponding bones; and CD-B2B evaluates the overall structural similarity between the predicted and actual skeletal bones. For skinning weights, we adopt Skin-L1 distance to directly measure the discrepancy between the predicted and ground-truth weight distributions.

Implementation. Our experiments run on a RTX 4090 GPU. The first stage trained for 2,000 epochs. The resulting coarse skinning weights were then utilized to train the subsequent model for joint estimation and refined skinning. The second stage trained for 300 epochs. The third stage trained 3,000 epochs.Training stages shared the same initial learning rate of 1e-5 and weight decay of 1e-5 with Adam optimizer

B. Comparisons

We follow the common evaluation protocol and compare our method with recent approaches, including RigNet [1] and the state-of-the-art method UniRig [2]. As for our approach, we separately take the network architectures of Point Cloud Transformer [24] and UniRig, forming two versions named Ours (PCT) and Ours (UniRig). All models are trained on the same training sets without data augmentation. The quantitative results are summarized in Table I. Comparing VRoid and VRoid+ datasets, the presence of diverse character poses leads to significantly higher prediction errors across all methods. Our approach achieves superior rigging accuracy, and this advantage becomes more evident on the VRoid+ dataset, indicating its stronger suitability for characters in non-standard poses. A consistent trend is observed on the Mixamo and Mixamo+ datasets, further validating the robustness of our method under pose variations.

Fig. 2 presents the predicted skeletons and skinning weights of these approaches, respectively.The first three rows show characters with diverse shapes but relatively standard poses. The skeletal predictions of RigNet and UniRig often exhibit deviations from the ground truth, placing joints near but outside the shape volume, while our approach, with the help of multi-skeleton fusion, corrects these deviations to obtain more accurate joint positions. The last three rows show characters with non-standard poses. Existing methods often predict distorted skeleton and accurate skinning weights, while ours obtains more reliable rigging results with the help of structure-aware encoding.

C. Ablation

The ablation experiments examine two key compositions, i.e. the Structure-Aware Encoding (SAE) in Stage2 and the Multi-Skeleton Fusion (MSF) in Stage3. The *baseline* refers to the setting without the two modules, equal to a single pass of the network used in our Stage2 without H_P as input. Then we separately remove the modules of the two stages, i.e. *w/o SAE* and *w/o MSF*, to demonstrate their effectiveness. For experiment *w/o SAE*, in Stage2, we adopt the FFT positional encoding different from Stage1 in the point cloud Transformer, so that the first two stages provide different estimations of the

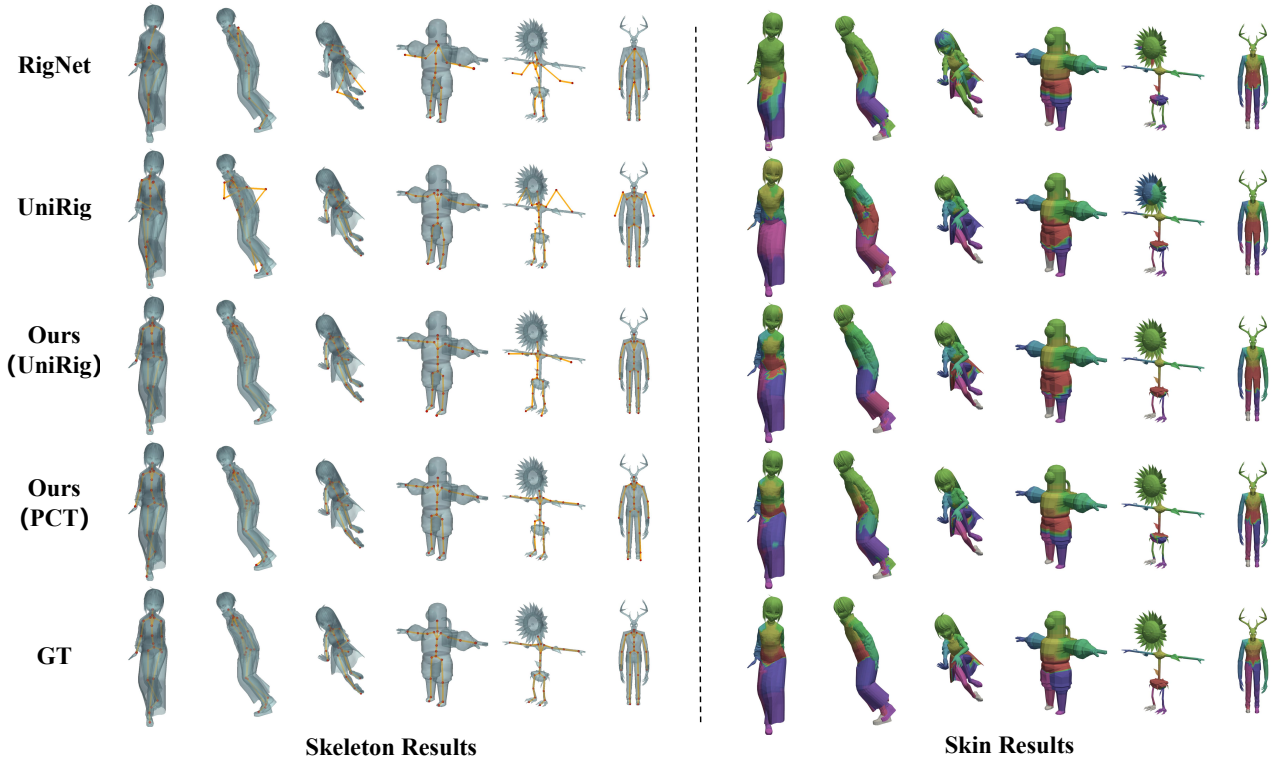

Fig. 2: **Comparative results of skeleton estimation and skinning**. Rows correspond to different methods and ground truth (GT). Left: Skeleton estimation results of different methods; Right: Skinning visualization results (different colors denote regions affected by distinct bones). The results of our proposed methods are more consistent with the ground truth in both skeleton structure and skinning weights.

skeletons and skinning weights neither without the structure-aware encoding, yet still support the multi-skeleton fusion in Stage3. For experiment *w/o MSF*, we simply remove the module of Stage3 and take the predictions from Stage2 as the final output. Table II reports the quantitative results of the baseline experiments. Since MSF in the third stage does not predict a new version of skinning weights, we mark the corresponding entries in the third row as '-'. The Structure-Aware Encoding (SAE) demonstrates its primary strength in significantly refining the estimation of skinning weights, as evidenced by its superior performance on the Skin-L1 metric across all datasets, leading to the further refinement of skeleton joints. Meanwhile, the Multi-Skeleton Fusion (MSF) module proves to be crucial for achieving the most accurate geometric correspondence, consistently delivering the lowest errors across all Chamfer Distance (CD) metrics (J2J, J2B, B2B). Fig. 3 also shows the superior performance of our approach improved by the two proposed modules.

V. Conclusion

Our paper presents a novel automatic rigging framework, SAEFRig, which progressively optimizes the prediction of skeleton joints and skinning weights through Structure-Aware Encoding and a Multi-Skeleton Fusion mechanism. Unlike traditional single-pass approaches, SAEFRig explicitly models the structural correspondence between shape and skeleton via Structure-Aware Encoding, and leverages Multi-Skeleton Fusion for fusion-based refinement, thereby significantly improving rigging quality under complex topologies and non-standard poses. Experimental results show that the method achieves state-of-the-art performance across multiple metrics, and ablation studies confirm the effectiveness of each module. SAEFRig provides a more robust solution for automatic rigging tasks, particularly suitable for animation preparation of AIGC-generated content, offering a solid step towards more intelligent and automated animation production.

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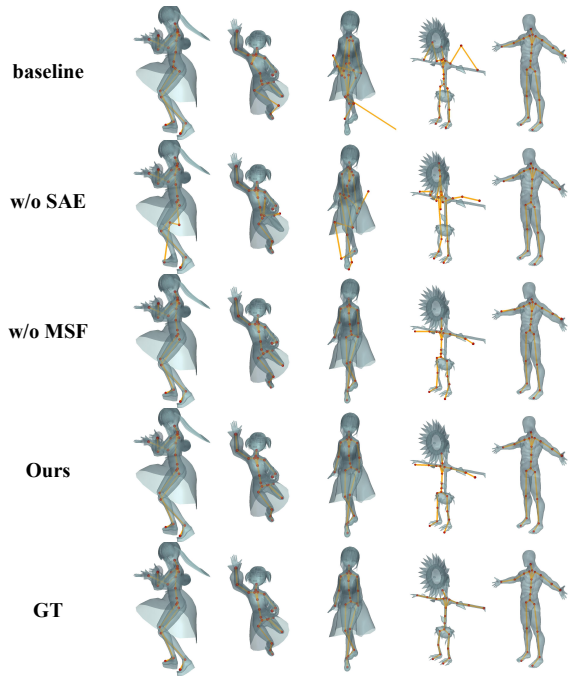

Fig. 3: **Skeleton Ablation Results**. Our full model generates the most accurate skeletal structure with reasonable joint positions, where SAE improves skeletal rationality and MSF refines joint placement.

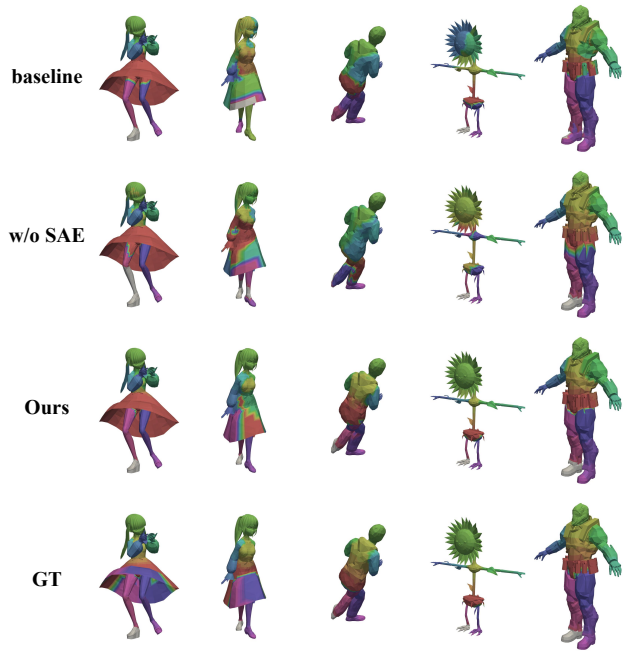

Fig. 4: **Skinning Ablation Results**. Since the Multi-Skeleton Fusion (MSF) module does not generate new skinning weights, we compare the weight distributions of the baseline, w/o SAE, and full model (Ours). Our full model generates the most accurate and smooth weight distributions.

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